

CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 18: Constructions

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Which angles are bisected to construct 75° ?
2. How many measurements are needed to construct (i) a quadrilateral (ii) a rectangle?
3. A quadrilateral has sides 4 cm, 5 cm, 6 cm, 4.5 cm and a diagonal 7 cm. What is drawn first?
4. A parallelogram has diagonals 10 cm and 8 cm. What lengths are marked from their point of intersection?
5. A rectangle has one side 8 cm and a diagonal 10 cm. What will the other side measure?
6. A rhombus has diagonals 12 cm and 16 cm. What will each side measure?
7. A square has a diagonal 8 cm. What will its side measure, to one decimal place?
8. In constructing a rhombus with side 5 cm and one angle 60° , what is the length of the shorter diagonal?
9. Can a quadrilateral be constructed with $AB = 3$ cm, $AD = 4$ cm and diagonal $BD = 8$ cm?
10. Write the steps to bisect a given angle.

Answer Key

1. The angle between 60° and 90°
2. 5 and 2
3. The diagonal, then a triangle on each side of it
4. 5 cm and 5 cm on one, 4 cm and 4 cm on the other
5. 6 cm
6. 10 cm
7. 5.7 cm
8. 5 cm
9. No; $3 + 4 < 8$, so triangle ABD is impossible
10. Cut both arms with an arc from the vertex; from the two points draw equal arcs to cross; join the vertex to the crossing point